HAMMAM OMER

Medical AI Researcher, Clinical Safety Auditor & Python Systems Architect

Omdurman, Sudan | GitHub: github.com/HammamNM | Audit Explorer: hammamnm.github.io/reasoning-debugger
Cloud Platform: hammam16.pythonanywhere.com | Preprint DOI: 10.5281/zenodo.23835945 | Blog: nexoramed.online

Executive Profile: Medical AI researcher and software builder specializing in Clinical LLM Safety Auditing, Neuro-Symbolic Verification, and Automated Reasoning Analysis. Architect of two open-source systems (Deterministic-First Hybrid Clinical Auditor & Cloud Clinical Reasoning Platform), backed by a DOI-published benchmark across 420 emergency clinical evaluations and technical articles on NexoraMed.

71.7% (38/53)	0 / 1,767	100% / 100%	2 LIVE TOOLS
JUDGE BLINDNESS EXPOSED	PARSER HALLUCINATIONS	CRITERION RECALL & SPEC.	CLOUD APP + TRACE EXPLORER

CORE ENGINEERING SYSTEMS & RESEARCH DELIVERABLES (2 LIVE TOOLS)

1. Deterministic-First Hybrid Clinical Auditor (v0.2-frozen-rules) [Interactive 420-Trace Explorer] | [GitHub Repo] | [50-Scenario Specs]

Lead Architect & Sole Author | Neuro-Symbolic Clinical Safety Verification & Interactive Audit Web App

- **Two-Stage Decoupled Architecture:** Isolated constrained semantic extraction (Stage I) from pure-Python deterministic clinical rule verification (Stage II, <2ms CPU latency) across 420 paired emergency management plans on 6 model ecosystems (gpt-oss-120b/20b, gemini-3.5/3.1-flash-lite, gemma-4-26b, qwen3.8-27b).
- **Exposed LLM-as-a-Judge Blindness:** Demonstrated empirically that standalone LLM judges missed 38 of 53 criterion-unsafe responses (71.7% false-pass rate; 95% Wilson CI: 58.4%-82.0%), including 100% of failures (27/27) produced by fluent flagship models under emergency pressure.
- **Resolved Lexical False-Alarm Collapse:** Eliminated the 48.2% false-alarm rate (177/367) of untyped keyword matchers via a canonical Assertion/Negation Gate (PROPOSED vs. NEGATED), achieving 100.0% specificity (367/367) and 100.0% criterion recall (Youden's J = 1.000).
- **Zero-Hallucination Substring Gate & Clinical Specs:** Enforced character-level grounding (norm(span) in norm(R)) with 0 hallucinations across 1,767 events across 50 emergency scenarios anchored to Class I AHA/ACC, ESC, BTS, EAST, and Surviving Sepsis guidelines.

2. Clinical Reasoning Platform & Domain-Independent Execution Kernel [Live Cloud Web App] | [Architecture Spec] | [Kernel Code]

Creator & System Architect | Cloud Web Application (hammam16.pythonanywhere.com) & Directed Graph Engine

- **Cloud Clinical Reasoning Platform:** Developed and deployed a live Python web application and execution kernel that transforms unstructured clinical decision flows and chain-of-thought traces into Directed Acyclic Reasoning Graphs (DAGs) with explicit provenance links.
- **Cognitive Bias & Structural Failure Detectors:** Implemented deterministic graph analyzers to localize epistemic failure patterns including anchoring bias, premature diagnostic closure, unsupported inference, and temporal prerequisite violations (idx(prereq) < idx(action)).
- **Production-Grade Contract Testing:** Enforced strict runtime boundary isolation, deterministic serialization round-trips, and zero-leakage contracts across 30+ automated pytest test suites.

PEER-CITABLE PREPRINT & TECHNICAL MEDICAL PUBLISHING

- **DOI Preprint (Zenodo / CERN):** Omer, H. (2026). Beyond Surface Fluency: A Deterministic-First Hybrid Clinical Auditor Exposes LLM-as-a-Judge Blindness and Behavioral Shifts Under Emergency Clinical Pressure. Official DOI: 10.5281/zenodo.23835945 (20 pages, 50 scenarios, 420 live evaluations).
- **Founder & Technical Author - NexoraMed (nexoramed.online):** Operating an English-language Medical AI publication focused on clinical reasoning frameworks, LLM safety evaluation, and artificial intelligence tools in medical education (2026 - Present).

TECHNICAL & CLINICAL ARSENAL

- **AI Safety & LLM Evaluation:** Multi-Ecosystem Benchmarking, LLM-as-a-Judge Failure Auditing, Sycophancy & Precautionary Shift Analysis, Automated Rubric & Protocol Engineering, Statistical Verification (Wilson 95% CIs, Exact McNemar Tests, 6-Tier Ablation).
- **Software & Systems Engineering:** Python (OOP, Type-Safe Data Models, Cloud Web Apps, CLI & REST APIs), Neuro-Symbolic Pipelines, Deterministic Substring & Regex Engines, JSON Schema Constrained Decoding, Pytest TDD, Git/GitHub, Linux/Termux CLI.
- **Clinical Domain Expertise:** Emergency Medicine Triage & Resuscitation Protocols, AHA/ACC, ESC, BTS & ATLS Guideline Interpretation, Differential Diagnosis Modeling, Contraindication & Temporal Prerequisite Auditing.

EDUCATION

- **Bachelor of Medicine, Bachelor of Surgery (MBBS Candidate - 3rd Year):** Faculty of Medicine, Omdurman Islamic University - Omdurman, Sudan (Ongoing, 2026).